20-04-2025

Agenda :—

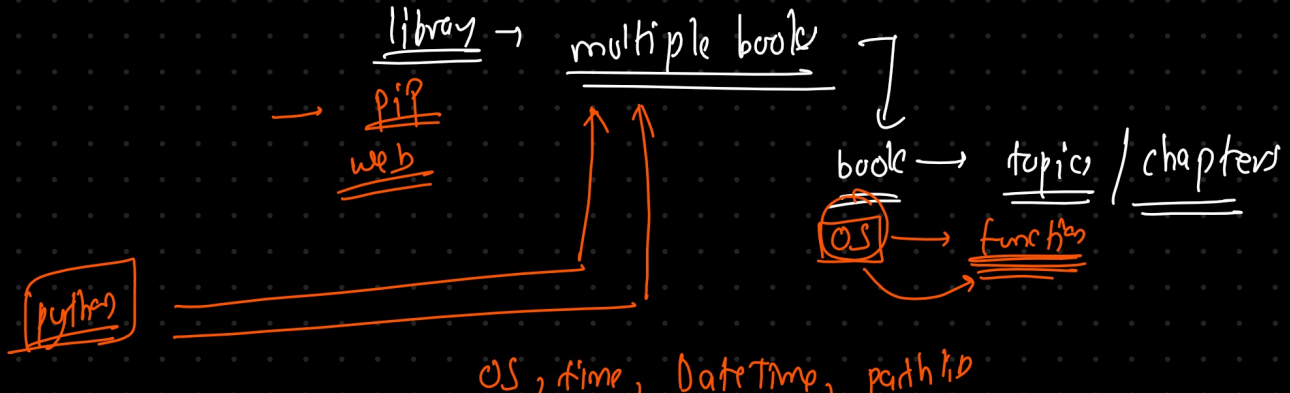
File operation & Exception

file → information / data

→ .xlsx .pdf , .txt , .csv , .jpg , .jpeg , .png .py

→ ram → harddrive

① ②



→ bmi → funch
→ x → get funct
→ y → get func

? → os / we → xyz

.ipynb

→ getcwd()
→ listdir()

Community → Data Sci

→ os
→ pandas
→ numpy
→ scikit → ML
→ matplot → Plotting
→ pytorch → calculations
→ cv2 → computer vision

safe to use open source library!

MNC]

xyz → ∅

os.listdir( )

→ getcwd( )

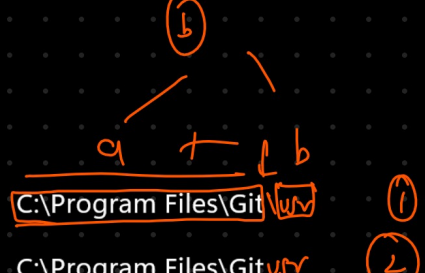
↓

all the file

os.listdir("path")

↓

path



① →

②

a+b

bin
cmd
etc
usr

a + " \ " + b →

os.path.join(a, b, c)

C:

User:

Mona:

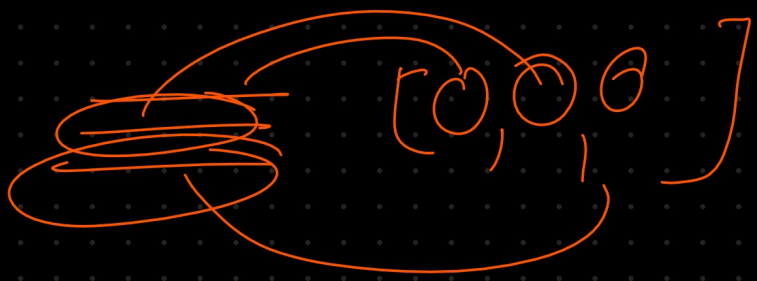
Test:

image / a.jpg → image/a.jpg → relative

a.ipynb

terminal → a.ipynb → relative → cur

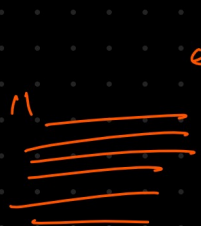
C:/User/mona/Test/a.ipynb → full path



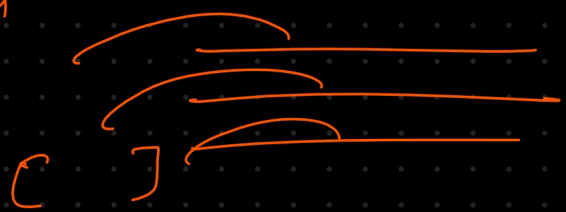
f = open(file-name)

f.read() → read all data

bring all data



freadlines() → read all data but line by line



freadline()

read → everything at once

readline() → single line

readlines() → single line but all lines as a list

Access()

open(file_path, mode) → "r"

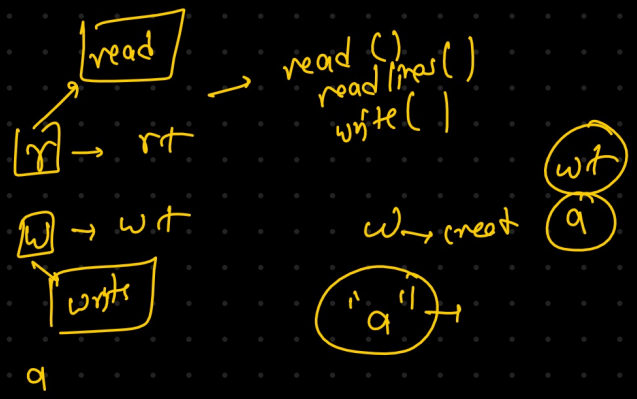
File modes:

"r" → Read, error if file does not exist (Default)

"w" → write, creates new or truncate existing file,

"a" → append,

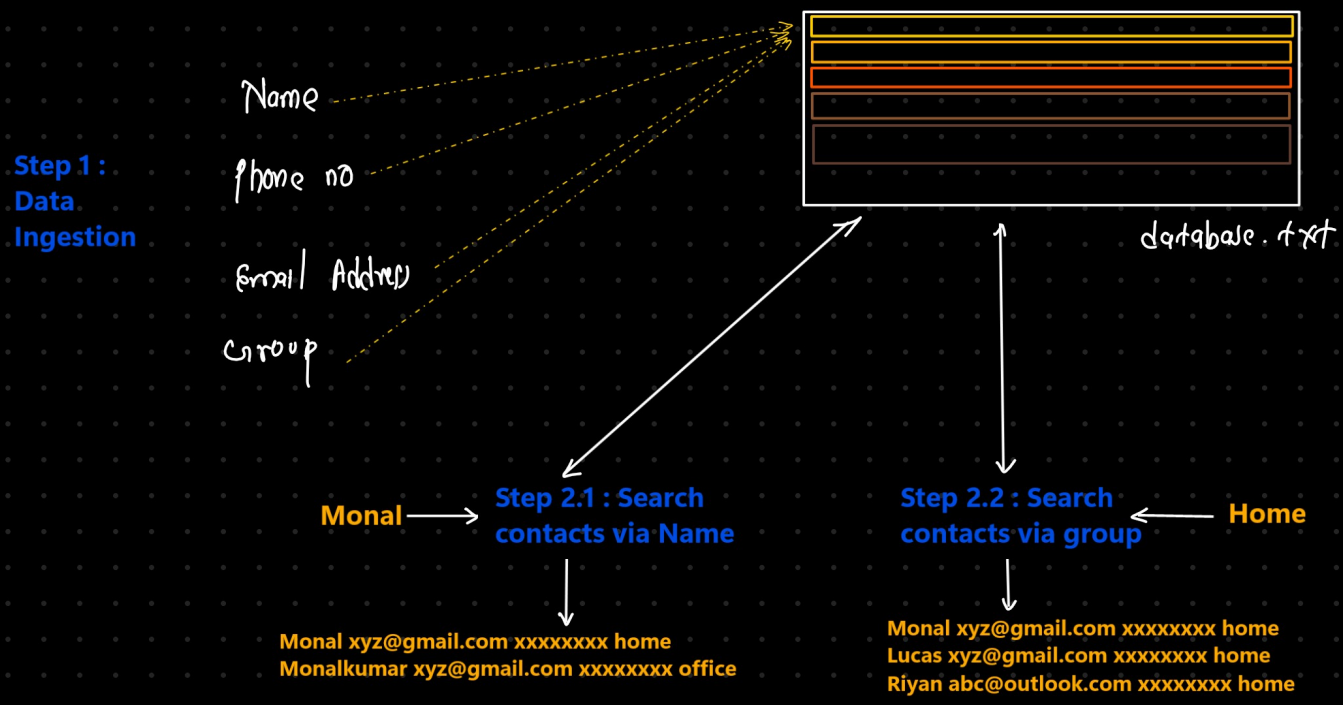
"+" → update



r+	read	write	file must exist	write from the beginning & does not truncate
w+	read	write	creates file if not exist	truncate/empty first then write
a+	read	write	create file	(appending from last line)

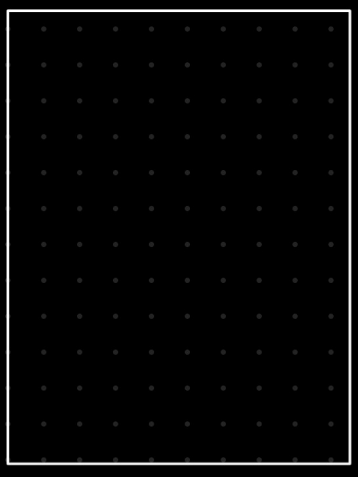
- **`r`**: Opens the file for reading only. Raises an error if the file doesn't exist.
- **`r+`**: Opens the file for both reading and writing. Raises an error if the file doesn't exist. Starts from beginning.
- **`w`**: Opens the file for writing only. Overwrites the file if it exists; creates a new file if it doesn't.
- **`w+`**: Opens the file for both writing and reading. Overwrites the file if it exists; creates a new file if it doesn't. Truncate the contents.
- **`a`**: Opens the file for appending data to the end. Creates a new file if it doesn't exist. Preserves the existing content.
- **`a+`**: Opens the file for both appending and reading. Creates a new file if it doesn't exist. Appends new data to the end while allowing reading from the beginning.

Assignment 1



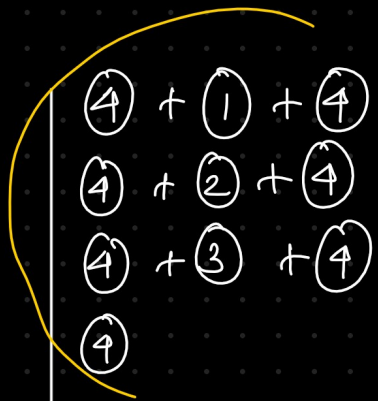
Step 2.3 : Ability to delete context (data should be removed from database.txt)

Assignment 2



To-do list

1. Add to task list
2. Remove task from list
3. Complete task from list
4. Get all active tasks (To-do)



Welcome to to-do list app :
Press 1 to add a new task
>> buy groceries
Press 2 to Remove task
>>Press id of task you want to remove
>> 2
Press 3 to complete task
>>Press id of task you want to mark as complete
>> 2
Press 4 to view to-do list
>>show all active task

Assignment 3 : Write a function to check whether number is palindrome.

Assignment 4 : Write a function which take total values as input and create Fibonacci series of that length

user : 5
output : 0, 1, 1, 2, 3

Assignment 5 : Write a function which take 2 list of same length and perform dot product.

user : [1,2], [1,2]
output : 5